Contiguous Nash territory division: the mathematical setup and the single-tree solver

Territory-division project*

2026-08-30

Abstract

Two merging sales forces must divide a set of postal territories (ZCTAs) between the two legacy representatives who overlap on each region. We state the model — maximum Nash welfare over indivisible zips at a zero disagreement point, with hard geographic contiguity — and develop the geometry that makes its computational behaviour intelligible: the feasible set is a finite point cloud in gain space, the free optimum is a hyperbola tangency, contiguity thins the cloud, and the solver's job is to prove that nothing above a hyperbola survives the thinning. We describe the two families of cutting planes (outer-approximation tangents for the concavified objective, root-free separator cuts for contiguity), why their convergence theories multiply in a multi-tree loop and compose in a single branch-and-cut tree, and the leading implementation, `scip_tree`. Two fully enumerable toy instances — ten zips on a path, sixteen on a grid — reproduce every phenomenon at a size where it can be drawn.

1 The problem

For each postal code (zip) z in a contested region there are three nonnegative numbers: firm-A sales A_z, firm-B sales B_z, and the market opportunity M_z. Two parameters translate them into what a territory is worth to each representative: the *transfer capture* $\theta \in [0, 1]$, the fraction of the other rep's book an inheriting rep retains, and the *headroom credit* $\lambda \in [0, 1]$, the rate at which untapped opportunity counts against booked production. With $c_1 = 1 - \lambda$ and $c_2 = \theta(1 - \lambda)$ (the *net* headroom convention),

$$u_a(z) \ = \ c_1 A_z + c_2 B_z + \lambda M_z, \qquad u_b(z) \ = \ c_2 A_z + c_1 B_z + \lambda M_z, \tag{1}$$

and the model requires pointwise nonnegative headroom, $M_z \geq \max(A_z + \theta B_z,\ B_z + \theta A_z)$. The reference values used throughout are $\theta = 0.40$, $\lambda = 0.30$.

An *allocation* is a subset $S \subseteq Z$ of the region's zips, read as "rep a gets S, rep b gets $Z \setminus S$." Utilities are additive, so each side's *gain* is a bundle utility:

$$g_a(S) = \sum_{z \in S} u_a(z), \qquad g_b(S) = \sum_{z \notin S} u_b(z). \tag{2}$$

*A self-contained companion to `nash_territory_division.tex` and the contiguity development programme (`research/contiguity/`). Solver numbers are from the harness runs of 2026-08-29/30 (`research/contiguity/RESULTS.md`); every number in the toy examples is computed by `toy_path.py` / `toy_grid.py` and injected as a macro, never transcribed.

The criterion. The allocation is chosen to maximise the Nash product,

$$\max_{S \subseteq Z} \; g_a(S) \cdot g_b(S), \tag{3}$$

i.e. *maximum Nash welfare* (MNW) — the Nash bargaining solution (Nash, 1950) at disagreement point $d = (0, 0)$. The zero disagreement point is a modelling decision with three grounds, in descending force. First, the threat point is not available: after the merger neither rep can revert to a legacy book, so the payoff to failed bargaining is zero, not the status quo ante. Second, at a zero baseline MNW over indivisible goods is Pareto efficient and envy-free up to one good (EF1) as a theorem (Caragiannis et al., 2019); subtracting constants from the utilities voids the guarantee. Third, it restores the correspondence with continuum Nash bargaining, whose solution at $d = 0$ is an optimal-transport map (Warren, 2025). It also removes a failure mode outright: gains (2) are positive by construction, so the bargaining set cannot be empty for any parameter values.

Census decomposition. Nationally there are many reps per firm. The *census* intersects the two legacy rep maps and decomposes the country into bilateral components: each contested component is a region Z with one A-rep and one B-rep, and (3) is solved independently per component. Production-size components run to 400–800 zips. (Dense many-on-many components need a multi-agent extension — component-level MNW $\sum_i \log g_i$ — outside this note's scope.)

Contiguity. The operational requirement that a rep's territory be drivable enters as a *hard constraint*: on the Rook-adjacency graph $G = (Z, E)$ of the region's zips, each side's allocation must induce a connected subgraph. Precisely, since the pair graph itself may have several connected components, the constraint is *component-wise*: inside each connected component K of G, both $S \cap K$ and $K \setminus S$ must be connected in $G[K]$ or empty. No compactness penalty appears in the objective ($\rho = 0$ is the model); a perimeter tie-break, when wanted, is a lexicographic post-pass, never a penalty. Connected fair division is NP-hard already for two agents with identical valuations (Deligkas et al., 2021), so an exact method with certificates — not a closed form — is the realistic target; see also Bouveret et al. (2017); Bilò et al. (2022) for the fair-division-on-graphs setting and Bei et al. (2022) for price-of-connectivity bounds (none are known for Nash welfare).

2 Geometry of the free problem

Everything about (3) becomes visual in *gain space*. Map every allocation to the point $\big(g_a(S), g_b(S)\big) \in \mathbb{R}^2_{\geq 0}$: the feasible set is a finite cloud of 2^n points (generically fewer, by coincidences), contained in the box $[0, G_a^{\max}] \times [0, G_b^{\max}]$ with $G_a^{\max} = \sum_z u_a(z)$, $G_b^{\max} = \sum_z u_b(z)$. Moving one zip z from b to a translates the point by $(+u_a(z), -u_b(z))$: the cloud is the Minkowski-type sum of n two-point sets, anchored at $(0, G_b^{\max})$. Figure 1 (left) shows the actual cloud of the $n = 10$ path toy of Section 6.

Three features of the picture do real work.

- **The Nash optimum is a hyperbola tangency.** Level sets of the objective are hyperbolas $g_a g_b = \text{const}$; the optimum is the point of the cloud touching the highest one. Because the level sets are strictly monotone, the optimum lies on the cloud's Pareto frontier — Pareto efficiency is by construction, and "equalising" criteria that minimise $|g_a - g_b|$ can leave the frontier entirely and hurt both sides at once; Nash cannot.

- **The frontier is near-linear, so weighted sums fail.** A scalarisation $w\, g_a + (1-w)\, g_b$ picks the frontier point supported by a line of slope $-w/(1-w)$. When the frontier hugs a line — which additive utilities with many small zips produce — almost every w selects a corner, and the interesting middle of the frontier is invisible to the method. The hyperbola, curved against the frontier's flatness, selects the interior point stably.

- **The fractional relaxation is the ratio sort.** Allow fractional shares $x_z \in [0,1]$. The relaxed feasible region in gain space is the zonotope spanned by the segments $\{x_z\,(u_a(z), -u_b(z))\}$, whose upper boundary is traced by sorting zips by the exchange rate $r_z = u_a(z)/u_b(z)$ and adding them in decreasing order. Maximising $g_a g_b$ over that boundary is the discrete-optimal-transport relaxation of the problem, and its value is an *upper bound* on (3). Rounding it — take the best prefix of the ratio order — is the "prefix heuristic."

Why the prefix rule is not a theorem. The classical threshold argument is marginal: at an optimum, no zip's transfer should help, and treating g_a, g_b as fixed while z moves gives a ratio threshold. But discretely the gains jump by $u_a(z), u_b(z)$ — finite amounts — and the argument only certifies a local optimum of the exchange neighbourhood, not of (3). Measured against exhaustive enumeration the best prefix is optimal only $\sim$45% of the time at any size ($n = 6$: optimal 40%, mean shortfall 3.91%, worst 53.9%; $n = 14$: 45%, 0.57%, 10.7%), though the shortfall vanishes with scale (mean 0.02% at $n = 50$, 0.0009% at $n = 400$). It is exact only for the utilitarian criterion $\{z : u_a(z) > u_b(z)\}$. The exact solver of Section 3 costs fractions of a second, so the prefix is used only as a preview and as a warm start.

3 Concavification and outer approximation

Write (3) with binaries $x_z \in \{0,1\}$ and the linear gains (2). The product $g_a g_b$ is not concave, but its logarithm

$$\log g_a + \log g_b \tag{4}$$

is concave *in the linear expressions* (g_a, g_b) — the transformation concavifies the objective rather than linearising it. Problem (3) is therefore a *convex* mixed-integer nonlinear program: maximise $z_a + z_b$ subject to $z_a \le \log g_a$, $z_b \le \log g_b$, integrality, and (later) contiguity.

Outer approximation (Duran and Grossmann, 1986; Fletcher and Leyffer, 1994) replaces each concave constraint by supporting hyperplanes. At a point $\hat{g} > 0$,

$$z \;\le\; \log \hat{g} + \frac{g - \hat{g}}{\hat{g}} \tag{5}$$

is valid for $z \le \log g$ and tight at $\hat{g}$. The algorithm alternates: solve the MILP with the current tangent set (its optimum is an *upper bound*, because the tangent set is a relaxation of the log constraint); evaluate the true objective at the MILP's allocation (a *lower bound*); add tangents (5) at the incumbent's gains; repeat. When the bounds meet, the incumbent carries an optimality *certificate*. Finiteness is the Duran–Grossmann argument: there are finitely many integer points, and a repeated integer point has its tangent already present, forcing the bounds together. On the free problem the loop needs 6–7 iterations regardless of n (validated against exhaustive enumeration on 120 instances per size: 100% match, bound gap identically zero; 0.10 s at $n = 50$, 0.84 s at $n = 400$). The ε-certified alternative — a piecewise-linear under-estimator of the log built once, to accuracy ε (Caragiannis et al., 2019; Vielma and Nemhauser, 2011) — trades the loop for a bounded a-priori error and is the basis of the `flow_pwl` cross-check.

4 Contiguity: separator cuts and why the engine matters

4.1 The cut family

Connectivity of an induced subgraph has two standard mixed-integer encodings: a compact one that ships flow from a root (Shirabe, 2005), and an exponential family of cutting planes generated lazily (Validi et al., 2021; Validi and Buchanan, 2022); one-shot comparisons are in Duque et al. (2011), and a linear-size planar alternative in Zhang et al. (2024). The programme's production choice is the cut family, in the *root-free, component-wise* form that the harness validated (any rooted form is unsound when the pair graph is disconnected — fixing a root inside one component forces growth toward a vertex other components cannot reach, and the resulting "dual bound" can sit strictly below feasible allocations the solver has already visited; this was observed, with a master value of 6.283 against a feasible 6.436, before the reformulation).

Let y denote a side ($y_z = x_z$ for side a; $y_z = 1 - x_z$ for side b). Suppose an integral point splits side y inside a pair component K into pieces, and let P be a piece that is not the largest. Choose

$$u = \arg\max_{z \in P} u_y(z), \qquad v = \arg\max_{z \in \text{largest piece}} u_y(z),$$

and let C be a *minimal* u,v-vertex-separator in $G[K]$, seeded at the neighbourhood $N_K(P)$ and greedily minimalised. The cut is

$$\sum_{w \in C} y_w \;\geq\; y_u + y_v - 1. \tag{6}$$

Validity: if a feasible allocation puts both u and v on side y, it must contain a y-path from u to v inside K; every u–v path meets C, so some $w \in C$ has $y_w = 1$ and the left side is ≥ 1, matching the right. If either endpoint is off the side, the right side is ≤ 0 and the cut is slack. The argument never mentions how u, v, C were chosen — *any* u,v-separator yields a valid inequality — which is what lets the same construction separate fractional LP points (threshold the LP values at t, split the support, same cut) without a new proof. No root is fixed anywhere, so the family is sound on disconnected pair graphs, and the bound it supports is global. Figure 2 (middle) draws one cut of exactly this form on the grid toy.

4.2 Two cut families, one tree

The full problem needs *both* families: tangents (5) for the objective and separators (6) for feasibility. Their finite-convergence arguments are different theories — outer approximation is Duran–Grossmann, lazy feasibility cuts are combinatorial Benders (Hooker and Ottosson, 2003; Codato and Fischetti, 2006) — and in a *multi-tree* loop (restart a fresh MILP after each round of cuts, the only architecture available on a solver without callbacks, such as HiGHS) the two iteration counts *multiply*: each tangent round can re-trigger every separation round. Worse, with $\rho = 0$ many allocations tie in the objective, the master jumps between distant near-ties, and cut generation thrashes — textbook Kelley instability (Lemaréchal et al., 1995), whose standard remedy is in-out stabilisation (Ben-Ameur and Neto, 2007; Fischetti and Salvagnin, 2010). This is the mechanism behind the legacy loop's non-convergence at scale, and it is structural, not a tuning problem.

The state of the art runs everything in *one* branch-and-cut tree (Quesada and Grossmann, 1992; Bonami et al., 2008; Kronqvist et al., 2018; Validi et al., 2021): a single enumeration in which both cut families are added lazily at nodes, every node inherits all cuts, and the tree's dual bound is globally valid at every moment. That is the architecture of the leading method.

4.3 What contiguity does to the geometry

In gain space, contiguity deletes points: the cloud of 2^n allocations thins to the connected ones, the Pareto frontier moves inward (weakly), and the tangency point slides down its hyperbola to a lower level — the *cost of contiguity*, measured as the relative drop in the Nash product. On the synthetic battery this cost is 0.03–5%, and on the harness tiers it settles around 0.4%; the EF1 property, a theorem only for the unconstrained optimum, has held empirically on every certified contiguous optimum the harness has produced. The toys of Section 6 make the thinning literal: on the path, $2^{10} = 1024$ points collapse to 56 intervals (Figure 1, right); on the grid, $2^{16} = 65536$ collapse to 1256 allocations with both sides connected.

5 The leading solver: `scip_tree`

`scip_tree` implements the single-tree architecture on SCIP via PySCIPOpt. The model, in the native formulation, is

$$
\begin{aligned}
\max\quad & z_a + z_b \\
\text{s.t.}\quad & g_a \le \textstyle\sum_z u_a(z)\, x_z, \qquad g_b \le \textstyle\sum_z u_b(z)\,(1 - x_z), \\
& z_a \le \log g_a, \qquad z_b \le \log g_b, \\
& \textstyle\sum_{z:\, u_a(z)>0} x_z \ge 1, \qquad \textstyle\sum_{z:\, u_b(z)>0} (1 - x_z) \ge 1, \\
& x \text{ contiguous component-wise (lazy cuts (6))}, \qquad x_z \in \{0,1\}.
\end{aligned} \tag{7}
$$

SCIP recognises $z \le \log g$ as convex, so its dual bound is a valid global upper bound on $\log g_a + \log g_b$ throughout the solve; an "OA" variant replaces the logs by lazily separated tangents (5) for the rungs where the nonlinear LP relaxations become numerically fragile. Contiguity lives in a constraint handler that (i) rejects and cuts off disconnected integral points with (6), (ii) *repairs* each rejected point (flip whole stray pieces, then a bounded boundary-swap search) and offers the repaired allocation back as an incumbent, and (iii) optionally separates fractional points at the root. Four engineering points carry the correctness and the performance; all four were found the hard way.

1. **Inequalities, not equalities, on the gains.** The objective is increasing in g_a, g_b through the logs, so $g_a \le \sum u_a x$ is tight at every optimum — but an equality lets presolve multi-aggregate the gain variables out of the transformed problem, after which no in-callback incumbent can be posted. Relatedly, dual presolve reductions must be off: they count variable locks over the constraints presolve can *see*, and the lazy rows are not there yet — with them on, presolve once fixed a tangent-bounded variable at its bound and "certified" a suboptimal warm start.

2. **The gain floor comes from the incumbent.** Any allocation at least as good as an incumbent of value L_0 has $g_a \geq e^{L_0}/\sum_z u_b(z)$; using that as g_a's lower bound (instead of an absolute 10^{-9}) takes the log's gradient at the bound from 10^9 to order 10^{-1} and is the difference between stable and unstable LPs at scale.

3. **A feasibility-tolerance ladder, keyed on the engine's stop reason.** SCIP's stored solution satisfies $z \leq \log g$ only to `feastol`, so a certificate recomputed from the solution is loose by $\approx$2·`feastol`; certifying to the harness tolerance 10^{-8} needs `feastol` $= 10^{-9}$. Small tolerances also make large LPs abort. The solver therefore descends a ladder — 10^{-9}, 10^{-7}, 10^{-6}, then the OA formulation — restarting each rung from the best allocation proved so far; every rung bounds the same optimum, so the merged answer keeps the best bounds across rungs. The retry decision keys on *whether the engine consumed its budget or crashed*, never on the harness-facing status: an LP abort surfaced as a polite `time_limit` once disabled the whole ladder silently, and every large instance stopped within a minute of a 1200 s budget. A bound from a rung at `feastol` f is a floating-point bound, rigorous to $O(f \cdot \|\text{duals}\|)$ — usually far tighter in practice, but a rigorous certificate at the two sizes now sitting exactly at the tolerance floor needs an exact post-hoc LP bound over the final cut set (planned as unit W6c).

4. **The primal side needs help.** SCIP's built-in heuristics are contiguity-blind, so almost nothing they produce survives the feasibility check. A spanning-tree bisection warm start (F1) is handed to the tree as a full solution, the in-callback repair recycles every rejected point, and — per the diagnostic below — an iterated local search around incumbents closes most of the remaining primal gap in seconds.

Where it stands empirically. On the screening tier (63 battery pairs, 30–82 zips), `scip_tree` certifies 62/63 at $\rho = 0$ with median time-to-certificate 0.08 s; every pair of ≤125 zips across the harness certifies in seconds. On the seven large pairs (124–464 zips, 1200 s cap) the picture after the ladder fix:

n	certified gap (nats)	time	reading
124	1.1×10^{-7}	219 s	tolerance floor, stops early
135	3.0×10^{-8}	280 s	tolerance floor, stops early
169	4.2×10^{-3}	1200 s	dual-bound-limited
197	4.2×10^{-3}	1200 s	dual-bound-limited
205	3.4×10^{-4}	1200 s	nearly closed
320	1.3×10^{-4}	1200 s	nearly closed
464	2.3×10^{-3}	1200 s	dual-bound-limited

A primal/dual diagnostic then separated the two remaining regimes. An iterated local search improves three of the five open incumbents within 1.4 s (by up to 1.6×10^{-3} nats) and then plateaus at values that two independent starts reproduce to 9–10 digits — almost certainly the optima. Warm-starting the tree *at* those allocations still fails to close the bound in 1200 s (169 and 197 hold $3\text{–}4 \times 10^{-3}$; 205 and 320 reach 1.35×10^{-4} and 2.4×10^{-5}). The residual gap at 160+ zips is therefore *dual*: the allocation is found in seconds; proving it is the expensive half, and the levers are stronger separation below the root, reductions that shrink the tree's node count, or accepting certified gaps of ≤0.4% of the Nash product at production size — with the allocation itself agreed by two independent methods.

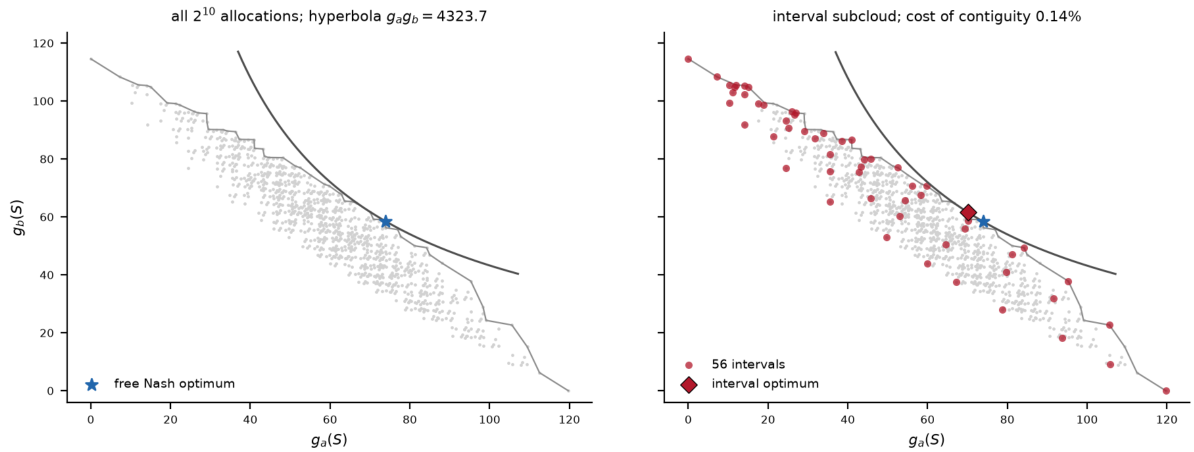

Figure 1: The path toy. *Left:* all $2^{10} = 1024$ allocations in gain space; the grey line is the Pareto frontier, the star the free Nash optimum $S^* = \{0, 1, 2, 3, 4, 6\}$ with $g_a g_b = 4323.7$, and the curve its hyperbola. *Right:* the 56 allocations in which side a's territory is an interval (red), their optimum $\{0, 1, 2, 3, 4, 5\}$ (diamond), and its lower tangent hyperbola. The cloud thins, the frontier retreats, the tangency slides.

6 Two toy instances

Both toys use the real utility model (1) at $\theta = 0.40$, $\lambda = 0.30$, and are small enough to enumerate completely, so every claim in this section is verified by brute force; the scripts regenerate the figures and every quoted number.

6.1 Ten zips on a path

Take $n = 10$ zips in a row (path graph), with A_z falling left to right, B_z rising, and lognormal noise on both (seed 0; M_z at 1.15× the headroom floor). This abstracts the common real geometry: two books strong on opposite ends of a region, contested in the middle.

The free optimum is $S^* = \{0, 1, 2, 3, 4, 6\}$: *not* an interval — it skips zip 5 and takes zip 6, because at these draws $u_a(6)/u_b(6)$ beats $u_a(5)/u_b(5)$ and the tangency prefers the exchange. Requiring side a to be connected restricts the 1024-point cloud to the 56 intervals ($\binom{10+1}{2} + 1$, verified by the enumeration); the constrained optimum is $\{0, 1, 2, 3, 4, 5\}$ with product 4317.6 against the free 4323.7 — a cost of contiguity of 0.14%. Requiring *both* sides connected is stronger still on a path (side a's interval must touch an end, else it cuts b in two): only 20 allocations survive, and here the optimum happens to be the same prefix $\{0, 1, 2, 3, 4, 5\}$, so the extra restriction costs nothing — an instance of a general pattern, that most of the cost is paid at the first connectivity requirement.

Both panels of Figure 1 repay staring. The frontier's near-linearity is why weighted-sum scalarisation collapses (Section 2); the vertical gap between the two hyperbolas *is* the cost of contiguity; and the interval subcloud hugs the free frontier closely — at production sizes the measured cost is a few tenths of a percent, and the toy shows why: connected near-optima crowd the tangency.

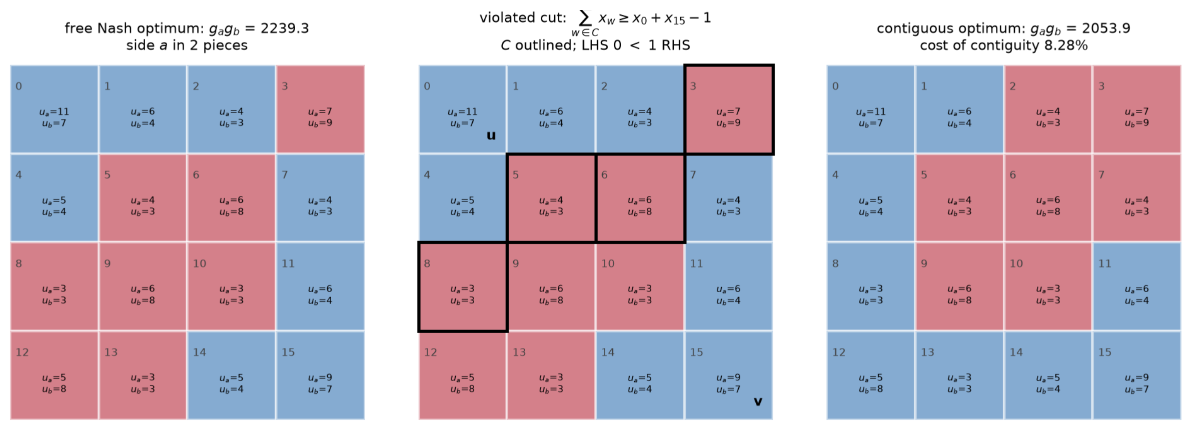

Figure 2: The grid toy (side a blue, side b red; per-cell utilities printed). *Left:* the free Nash optimum puts side a's territory in 2 pieces. *Middle:* the root-free separator cut (6) generated at that point: $P = \{0, 1, 2, 4\}$, endpoints $u = 0$, $v = 15$, separator $C = \{3, 5, 6, 8\}$ (outlined); the free point has $\sum_{w \in C} x_w = 0 < 1 = x_0 + x_{15} - 1$. *Right:* the contiguous optimum buys connectivity through the band.

6.2 Sixteen zips on a grid: disconnection and one cut

The path cannot show the failure mode that matters, because on a path the free optimum rarely fragments badly. Take a 4×4 Rook grid (seed 0) with a *bimodal* profile for firm A — two high-A pockets in opposite corners — and a high-B band on the anti-diagonal between them. This abstracts failure mechanism *(a)* of the programme: the free optimum wants both pockets and drops the glue.

The free optimum ($g_a g_b = 2239.3$) gives a the set $\{0, 1, 2, 4, 7, 11, 14, 15\}$ — 2 pieces, one per pocket. Of the $2^{16} = 65536$ allocations only 1256 have both sides connected; the best of them ($\{0, 1, 4, 8, 11, 12, 13, 14, 15\}$, $g_a g_b = 2053.9$) keeps one pocket, crosses the band at its cheapest cell, and reaches the far pocket along the edge — at a cost of 8.28%, an order of magnitude above the path toy's, because here contiguity binds against the value structure rather than against noise. (The toy is engineered to make the cost visible; measured production costs sit near 0.4%.)

The middle panel shows the solver's actual move at the free point. The smaller piece is $P = \{0, 1, 2, 4\}$; the highest-utility vertices give $u = 0$, $v = 15$; the neighbourhood of P, greedily minimalised, gives the separator $C = \{3, 5, 6, 8\}$. The free point violates (6) ($0 < 1$), the contiguous optimum satisfies it, and — the point of the root-free form — the inequality says nothing about *where* side a must live, only that it cannot occupy both u and v without paying for a crossing of C. Branch-and-cut adds such inequalities lazily, each one deleting a slab of the disconnected cloud, until the surviving relaxation's bound meets an incumbent like the right panel.

What the toys do not show

Honesty about the abstraction: (i) at $n \leq 16$ the certificates are instant — the dual-bound crawl of Section 5 needs hundreds of zips and value concentration to appear; (ii) real pair graphs arrive already disconnected (islands), which is what forces the component-wise statement of the constraint and the root-free cut form; (iii) real instances carry hundreds of near-zero "glue" zips whose only role is connectivity, a regime the reduction layer targets and no 16-cell toy can imitate.

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